



# 使用StreamReader和StreamWriter

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# 概述



对于文本文件来说，按字符读取，需要关注的一个问题就是字符集。



JDK提供了InputStreamReader或OutputSteamWriter两个组件，可以使用指定的字符集读取或写入文本文件。

```
public InputStreamReader( InputStream in, String charsetName)
```

```
public OutputStreamWriter( OutputStream out, String charsetName)
```

# 按照特定编码读取与写入文本文件



```
FileOutputStream fos = new FileOutputStream("polish.txt");  
OutputStreamWriter osw = new OutputStreamWriter(fos, "8859_2");  
char ch = '\u0323'; // Accented N.  
osw.write(ch);
```

读与写必须使用  
相同的字符集

```
FileInputStream fis = new FileInputStream("polish.txt");  
InputStreamReader isr = new InputStreamReader(fis, "8859_2");  
char ch = isr.read(ch);
```



读写字符时采用不同的字符集，将会导致乱码的出现。

# 文件文件读写示例



```
private static void writeToFile() {  
    try (var writer = new OutputStreamWriter(  
        new FileOutputStream(fileName),  
        StandardCharsets.UTF_8)) {  
        writer.write( str: "白日依山尽\n");  
        writer.write( str: "黄河入海流\n");  
        writer.write( str: "欲穷千里目\n");  
        writer.write( str: "更上一层楼\n");  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
    System.out.println("写入结束");  
}
```

```
private static void readFromFile() {  
    try (var reader = new InputStreamReader(  
        new FileInputStream(fileName),  
        StandardCharsets.UTF_8)) {  
        //一个字符一个字符地读  
        int value;  
        while ((value = reader.read()) != -1) {  
            System.out.print((char) value);  
        }  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}
```

UseInputStreamReaderWriter.java

# 读取文本文件



如果要使用当前系统的**默认字符集**来读取或写入文本文件，可以使用 `FileReader` 和 `FileWriter` 两个类：

```
FileWriter out = new FileWriter("output.txt");  
out.write(...);
```



相当于

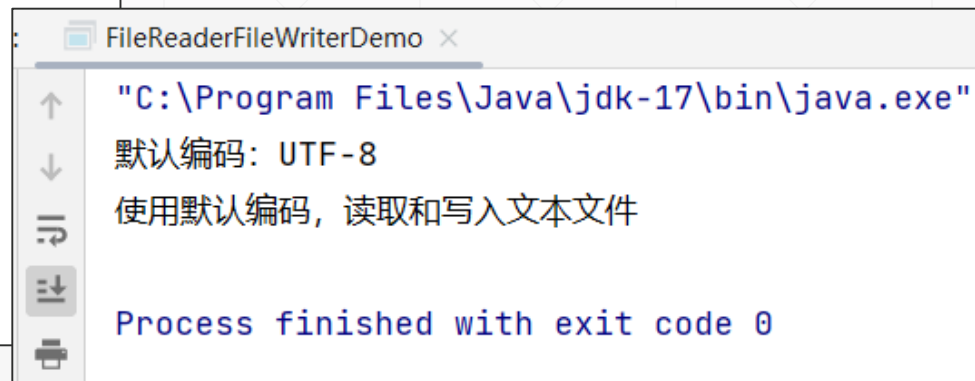
```
var fos = new FileOutputStream("output.txt");  
var osw = new OutputStreamWriter(fos,  
    Charset.defaultCharset());
```

# FileReaderWriter示例

```
public class FileReaderFileWriterDemo {  
    final static String MSG = "使用默认编码, 读取和写入文本文件";  
    final static String FILE_NAME = "temp.txt";  
    public static void main(String[] args) throws IOException {  
        System.out.println("默认编码: " + Charset.defaultCharset());  
        //写入文件  
        try (FileWriter fw = new FileWriter(FILE_NAME)) {  
            fw.write(MSG, off: 0, MSG.length());  
        }  
        //读取文件  
        char[] buf = new char[MSG.length()];  
        try (FileReader fr = new FileReader(FILE_NAME)) {  
            fr.read(buf, off: 0, MSG.length());  
            System.out.println(buf);  
        }  
    }  
}
```

FileReaderFileWriterDemo.java

运行截图:



```
FileReaderFileWriterDemo x  
"C:\Program Files\Java\jdk-17\bin\java.exe"  
默认编码: UTF-8  
使用默认编码, 读取和写入文本文件  
  
Process finished with exit code 0
```